

Deep learning NLP

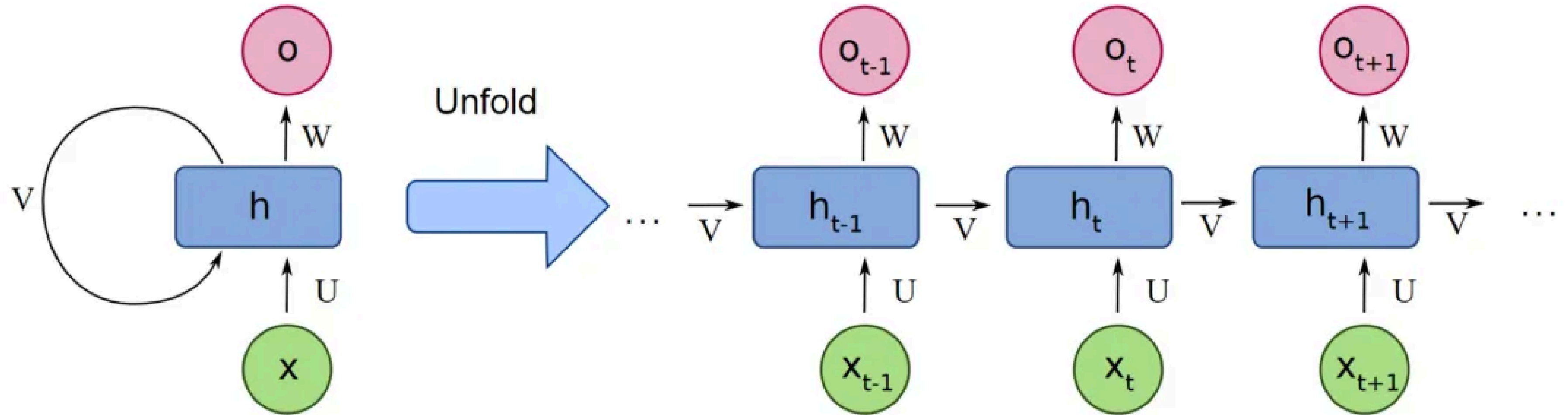
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1- RNN

An RNN (Recurrent Neural Network) is a type of neural network designed to work with sequential data — where the order of data matters, such as in text, time series, speech, or video frames.

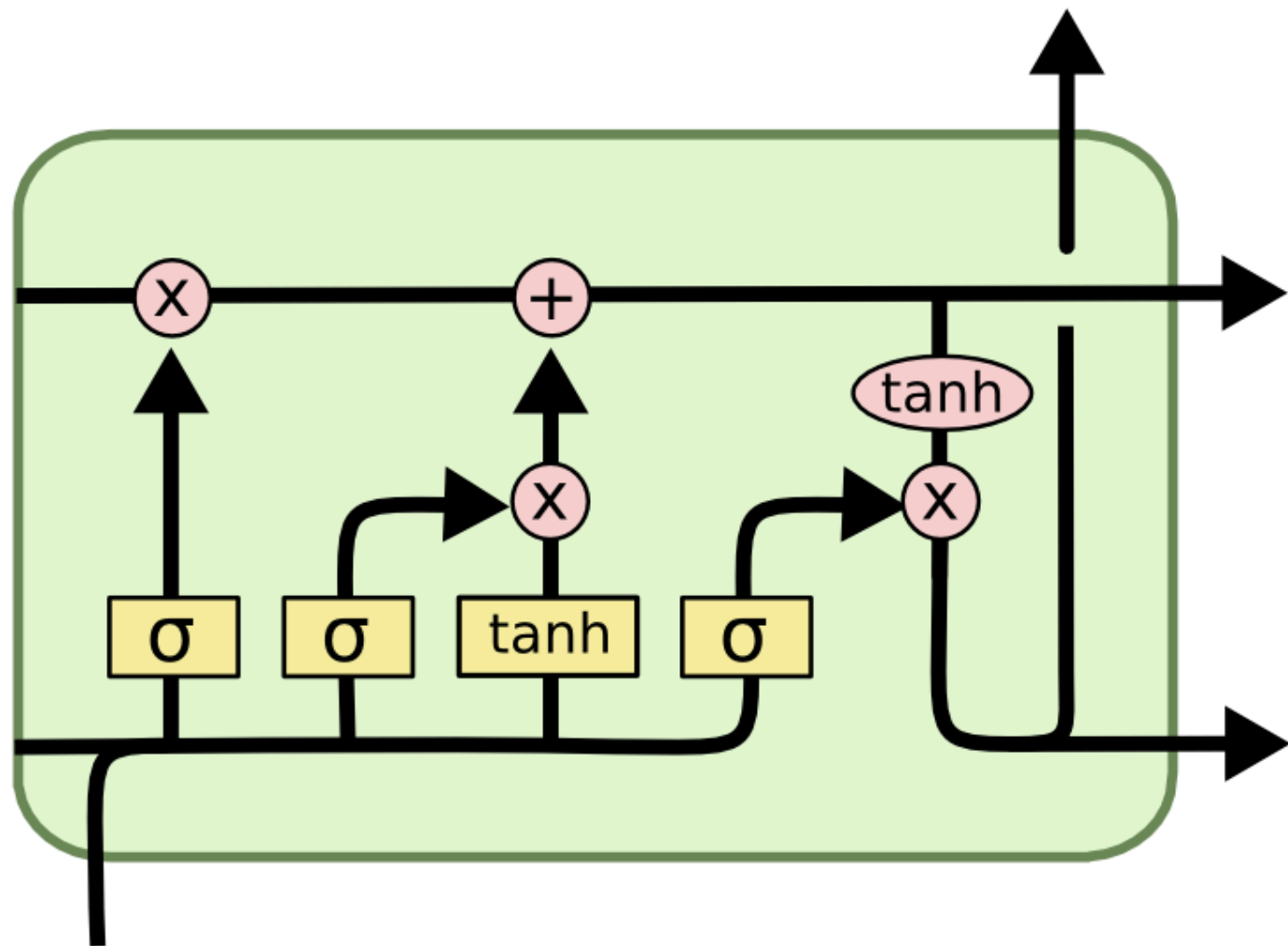
RNNs maintain a hidden state (memory) that gets updated at each step in the sequence, allowing the network to remember information from previous time steps.



2- LSTM

An LSTM (Long Short-Term Memory) network is an improved version of a vanilla RNN that overcomes the vanishing gradient problem by introducing a special memory cell and gates that control information flow. It was designed to capture long-term dependencies in sequential data.

LSTM introduces a cell state that runs through the sequence with minor linear interactions, helping preserve gradients and memory across long sequences.



Summary Table

Component	Purpose	Activation
Forget gate f_t	Remove info from memory	Sigmoid
Input gate i_t	Decide what new info to store	Sigmoid
Candidate $\tilde{c}_t$	Create new candidate values	Tanh
Cell state c_t	Long-term memory	Linear + Element-wise ops
Output gate o_t	Filter output from memory	Sigmoid
Hidden state h_t	Final output	Tanh

3- bidirectional RNN

